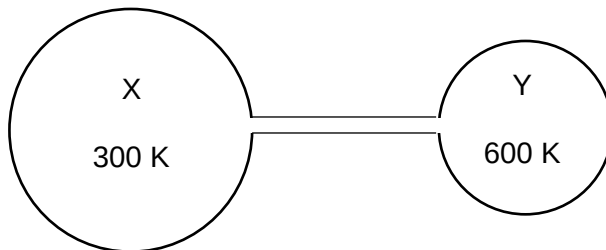


- 21 In the diagram, the volume of bulb X is twice that of bulb Y. The system is filled with an ideal gas and a steady state is established with the bulbs held at 300 K and 600 K.



There is a total of n moles of gas in the bulbs.

How many moles of gas are in bulb X?

A $\frac{1}{3}n$

B $\frac{1}{2}n$

C $\frac{2}{3}n$

D $\frac{4}{5}n$

21	D	$n = \frac{pV}{RT}$ So total number of moles in the bulbs $n = \frac{p(2V)}{R(300)} (\text{bulb X}) + \frac{pV}{R(600)} (\text{bulb Y})$ note that the pressure is the same in both bulbs $n = \frac{pV}{R} \left(\frac{1}{150} + \frac{1}{600} \right) = \frac{1}{120} \frac{pV}{R}$ $p = \frac{120nR}{V}$ So the number of moles in X $= \frac{p(2V)}{R(300)} = \frac{120nR}{V} \times \frac{2V}{300R} = 0.8 n$
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[Turn over

22 In a constant flow determination of the specific heat capacity of a liquid, it is important to